Lecture 2 2026-09-03

Today: linear algebra "crash course", gradient descent

Vector norms: assign "size" to a vector that we will later want to minimize

properties: $\|x\|_p$ (p-norm)

1. $\|x\|_p \geq 0 \quad \forall\, x \in \mathbb{R}^n$
   and $\|x\|_p = 0$ if and only if $x = 0$

2. $\|cx\|_p = c\,\|x\|_p$ for $c \in \mathbb{R}$

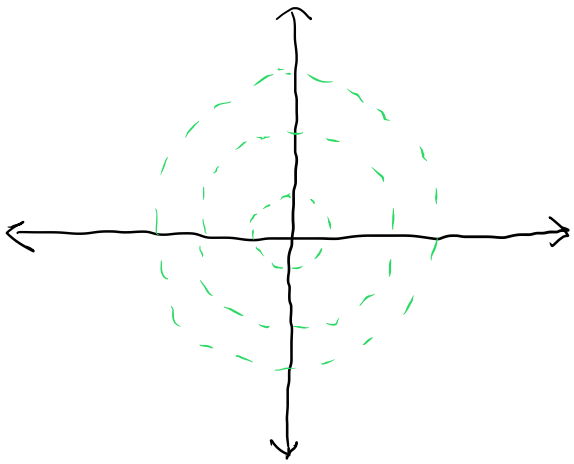
3. $\|x+y\|_p \leq \|x\|_p + \|y\|_p$

p-norm $\|x\|_p = \left(\sum_{i=1}^{n} |x_i|^p\right)^{1/p}$

2-norm: $\|x\|_2 = \sqrt{\sum_{i=1}^{n} (x_i)^2}$

colloquially: "as the crow flies"

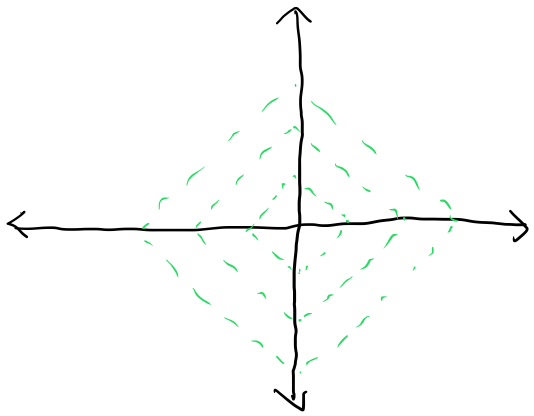
level sets look like:



1-norm: $\|x\|_1 = \sum_{i=1}^{n} |x_i|$

colloquially: Manhattan distance or
"taxi cab" norm

level curves look like:



l_1-norm is known as "sparsity promoting"

i.e., $\underset{x}{\text{minimize}} \; \|x\|_1$

often yields more 0-elements in the optimal solution.

→ useful property in e.g., signal processing, ML

Property: if p, q are integers such that $0 < p < q$, then

$$\|x\|_p \geq \|x\|_q \text{ for all } x$$

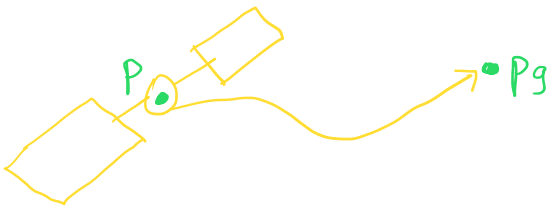
Why do we care?

Later on, we'll use the $\|x\|_p$ norm to minimize useful physical properties

e.g., if a spacecraft position is described as $p = (p_x, p_y) \in \mathbb{R}^2$, then

$$\underset{p}{\text{minimize}} \; \|p - p_g\|_p$$

will minimize position with respect to some desired goal p_g



Eigenvalues

If A is a square matrix

$A \in \mathbb{R}^{n \times n}$, then eigenvalue λ

$$Ax = \lambda v$$

Solve for eigenvalue via its characteristic equation:

$$\det | A - \lambda I | = 0$$

Eigenvalue decomposition

Let $v_1, \dots, v_n$ be n linearly independent eigenvectors of A.

Then we can express $A = T \Lambda T^{-1}$

where $T = \begin{pmatrix} v_1 & v_2 & \dots & v_n \end{pmatrix}$

$$\Lambda = \begin{pmatrix} \lambda_1 & & & 0 \\ & \lambda_2 & & \\ & & \ddots & \\ 0 & & & \lambda_n \end{pmatrix}$$

Computational complexity

How does the runtime of an algorithm scale with respect to the size n with respect to its argument/input?

→ The choice and performance of the optimization algorithm we will use is determined by computational complexity

common to consider big-O notation:

$\mathcal{O}(\cdot)$ input size n

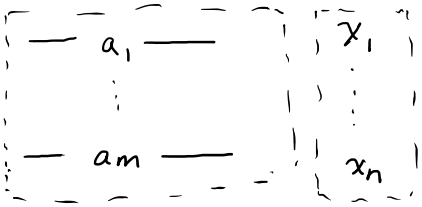
examples:

$\mathcal{O}(n)$: linear time

adding two vectors of size n

```
def add(x,y):
    sum = (0,...,0)
    for i=1→n
        sum_i = x_i + y_i
    return sum
```

matrix multiplication $y = Ax$ $A \in \mathbb{R}^{m \times n}$




```
def multiply(A, x):
    y = (0, ..., 0)
    for i = 1 → m
        y_i = 0
        for j = 1 → n
            y_i += A_ij x_j
    return y
```

Matrix inverse: A is square and invertible

$A \in \mathbb{R}^{n \times n}$

→ complexity of A^{-1} is $O(n^3)$

→ "optimized" routines used in high-performance linear algebra libraries get down to $O(n^\omega)$ where $\omega \approx 2.40$

Gradient descent

We will study problems of the form:

$$\min_{x \in \mathbb{R}^n} \quad f(x)$$

$$\text{subj. to:} \quad f_i(x) \leq 0 \qquad i = 1, \ldots, n_{ineq}$$

$$h_i(x) = 0 \qquad i = 1, \ldots, n_{eq}$$

Gradient:

if $f(x)$ such that $f: \mathbb{R}^n \to \mathbb{R}$

and $f \in \mathcal{C}^1$

$\rightarrow$ gradient $\nabla f(x) = \begin{pmatrix} \partial f / \partial x_1 \\ \vdots \\ \partial f / \partial x_n \end{pmatrix}$

Hessian:

if $f: \mathbb{R}^n \to \mathbb{R}$ and $f \in \mathcal{C}^2$

then Hessian $\nabla_{xx} f(x) = \begin{pmatrix} \partial^2 f/\partial x_1^2 & \cdots & \partial^2 f/\partial x_1 \partial x_n \\ \vdots & \ddots & \vdots \\ \partial^2 f/\partial x_n \partial x_1 & \cdots & \partial^2 f/\partial x_n^2 \end{pmatrix}$

H is a symmetric matrix and $H \in \mathbb{R}^{n \times n}$

The gradients and Hessians will be useful when we want to construct a local approximation of $f(x)$ using Taylor series

recall scalar version when $f: \mathbb{R} \to \mathbb{R}$

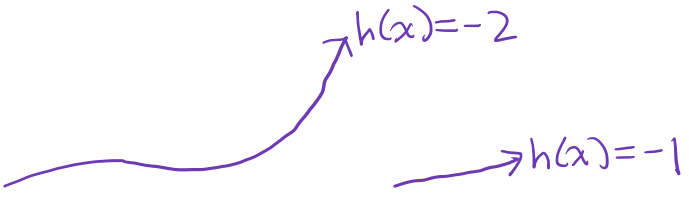
$$\to f(x) \approx f(\bar{x}) + f'(\bar{x})(x-\bar{x}) + \frac{1}{2} f''(x)(x-\bar{x})^2$$

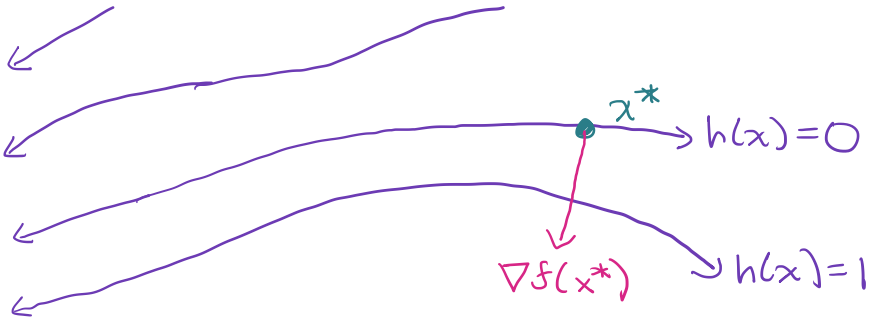
in vector version for $f: \mathbb{R}^n \to \mathbb{R}$

$$\to f(x) \approx f(\bar{x}) + \nabla f(\bar{x})^T (x-\bar{x}) + \frac{1}{2} (x-\bar{x})^T \nabla_{xx} f(\bar{x}) (x-\bar{x})$$

What information about a function does the gradient $\nabla f(x)$ provide?

Consider the level curves of some function $f(x)$.





A gradient points orthogonal to the level curve and in the direction of increasing value (low-to-high)

Key insight: if we want to minimize $f(x)$, then $\nabla f(x)$ can provide us information about direction of increase (and hence decrease)